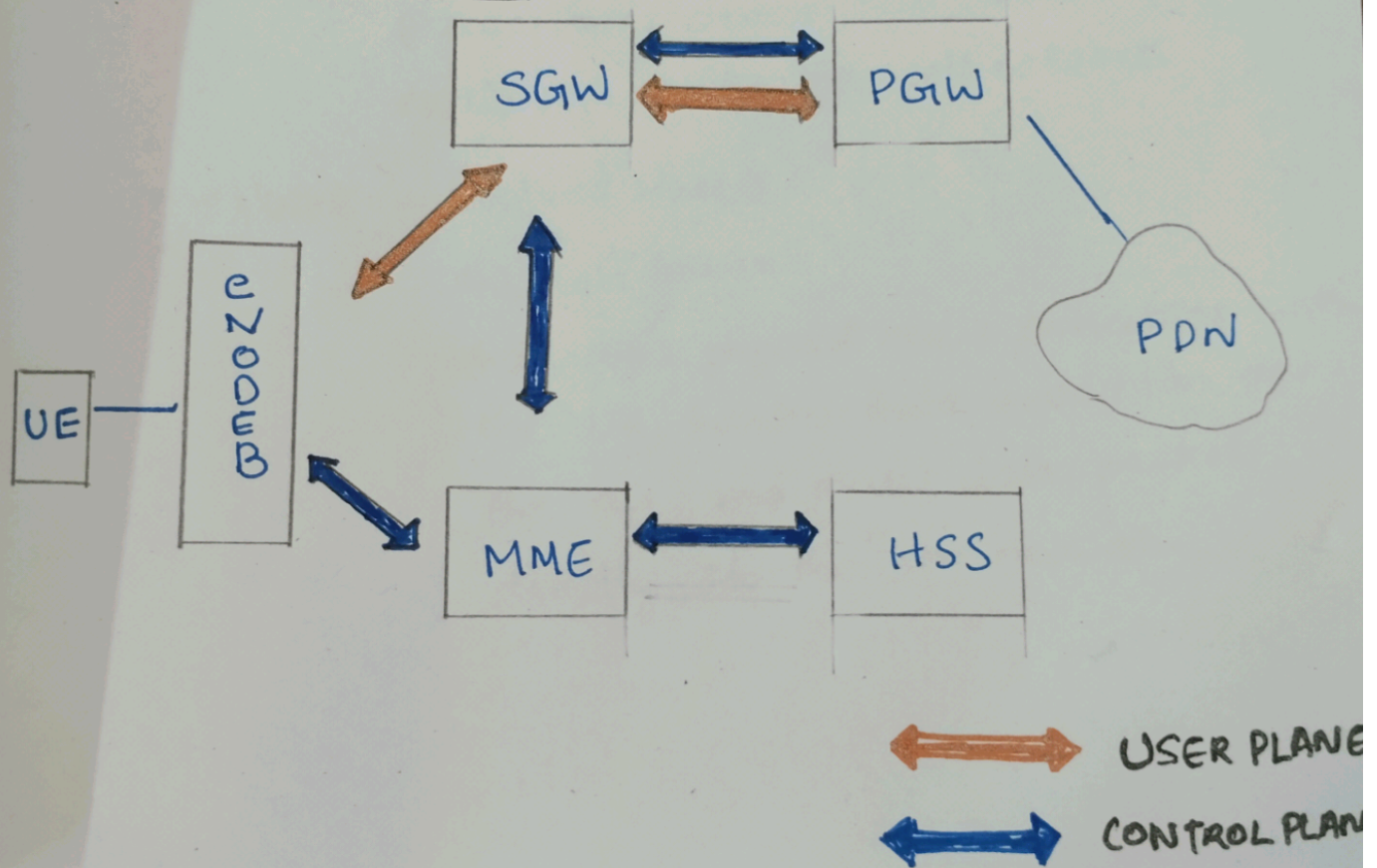


LTE Network Architecture

LTE Network Nodes



1. The Radio Network (E-UTRAN)

- UE - USER EQUIPMENT
 - Subscriber's device that connects via radio signals to the cell network.
- eNodeB - Evolved Node B
 - 4G cell tower
 - manages physical radio connections with the UE, handles data encryption over the air, and routes data packets directly into the core network.

2. The Control Plane (The Coordinators)

- MME — Mobile Management Entity
 - Brain of the core network control plane.
 - doesn't handle actual internet traffic.
 - It manages user authentication via HSS, tracks device location, and coordinates handovers between cell towers.
- HSS — Home Subscriber Server
 - Master database.
 - stores user subscription profiles, sim card authentication keys and

tracks which MME the user is currently assigned to.

3. THE USER PLANE (The Data Traffic Handlers)

- SGW - Serving Gateway
 - It acts as a mobility anchor, forwarding internet data packets back and forth between the eNodeB and the P-GW as the user moves.
- PGW - Packet Data Network Gateway
 - The exit/entry point to the outside world.

- It connects the LTE N/w to external IP networks (internet or IMS for VoLTE).
- It assigns IP addresses to UEs, enforces data filtering and coordinates closely with billing systems (OCS/PCRF) to track data usage.

Note:- The nodes use the control plane, in order to exchange signalling messages between each other. The signalling message controls the data session of the UE.

- Nodes use the user plane for exchange of exact data.